# Scientific Machine Learning for the Double Pendulum: A Comparative Study of Structure-Preserving Neural Networks

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## 1 Introduction

The double pendulum is a paradigmatic chaotic system in classical mechanics. Despite having only two degrees of freedom, it exhibits extreme sensitivity to initial conditions, complex phase-space topology, and positive Lyapunov exponents at moderate energies. These properties make it an ideal benchmark for evaluating Scientific Machine Learning (SciML) methods, which aim to integrate physical priors into data-driven models.

We present a systematic comparison of six learning approaches applied to the double pendulum: a baseline MLP, a Residual Network (ResNet), Neural ODE [**?**], Hamiltonian Neural Network (HNN) [**?**], Lagrangian Neural Network (LNN) [**?**], and a Physics-Informed Neural Network (PINN) [**?**]. We also compare four numerical integrators for generating reference solutions. Our focus is on two key metrics: *trajectory accuracy* (short- and long-term MSE) and *energy conservation* (drift of the Hamiltonian).

## 2 Methods

### 2.1 Mathematical Formulation

The double pendulum consists of two point masses $m_1, m_2$ connected by rigid rods of lengths $l_1, l_2$. The Lagrangian is $\mathcal{L} = T - V$ where

$$T = \tfrac{1}{2}m_1 l_1^2\dot{\theta}_1^2 + \tfrac{1}{2}m_2\big[l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)\big] \quad (1)$$

$$V = -(m_1+m_2)gl_1\cos\theta_1 - m_2gl_2\cos\theta_2 \quad (2)$$

The Euler-Lagrange equations yield coupled second-order nonlinear ODEs, which we rewrite as a first-order system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y})$ with state $\mathbf{y} = [\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2]^\top$. The Hamiltonian $H = T + V$ is conserved.

### 2.2 Numerical Solvers

We compare four integrators: (i) adaptive RK45 (reference, tol $= 10^{-10}$), (ii) fixed-step RK4, (iii) Symplectic Euler, and (iv) Störmer-Verlet (leapfrog). Symplectic methods operate in Hamiltonian $(q, p)$ coordinates and preserve the symplectic two-form, yielding bounded energy error.

### 2.3 Machine Learning Models

All models learn the map $\mathbf{y} \to \dot{\mathbf{y}}$ from 200 training trajectories (random ICs, $t \in [0, 10]$, $\Delta t = 0.01$).

**Baseline MLP:** 3-layer network with tanh activation; no physics prior.

**ResNet:** 4 residual blocks; can be interpreted as a discretized ODE ($\mathbf{y}_{n+1} = \mathbf{y}_n + f(\mathbf{y}_n)$).

**Neural ODE:** Learns continuous dynamics $\dot{\mathbf{y}} = f_\theta(\mathbf{y})$ and integrates with a differentiable solver (dopri5) [**?**].

**HNN:** Learns a scalar Hamiltonian $\hat{H}(\mathbf{q}, \mathbf{p})$ and derives dynamics via $\dot{q}_i = \partial\hat{H}/\partial p_i$, $\dot{p}_i = -\partial\hat{H}/\partial q_i$ [**?**]. The learned $\hat{H}$ is automatically conserved.

**LNN:** Learns a scalar Lagrangian $\hat{\mathcal{L}}(\mathbf{q}, \dot{\mathbf{q}})$ and obtains accelerations via the Euler-Lagrange equation, requiring the Hessian $\partial^2\hat{\mathcal{L}}/\partial\dot{\mathbf{q}}^2$ [**?**].

**PINN:** Maps time directly to state; trained with a combined data loss and physics residual loss that enforces the governing ODE at collocation points [**?**].

## 3 Results

### 3.1 Numerical Solver Comparison

RK45 (adaptive) achieves energy errors of $\mathcal{O}(10^{-8})$ over 50 s. Fixed-step RK4 shows $\mathcal{O}(10^{-4})$ drift at $\Delta t = 0.01$. The Störmer-Verlet integrator, while only 2nd-order accurate, exhibits bounded oscillatory energy error (no secular drift), confirming its symplectic structure. Convergence tests show RK4 achieves 4th-order convergence, while Symplectic Euler is 1st-order.

### 3.2 Chaos Analysis

We estimate the maximal Lyapunov exponent $\lambda_{\max} \approx 1.5\,\mathrm{s}^{-1}$ at $(\theta_1, \theta_2) = (\pi/2, \pi/2)$, confirming chaotic dynamics. Perturbations of $10^{-6}$ diverge to $\mathcal{O}(1)$ within 10 s. The Poincaré section at $\theta_1 = 0$ reveals the characteristic mixed regular-chaotic structure.

### 3.3 ML Model Comparison

Table **??** summarizes the results. Structure-preserving networks (HNN, LNN) achieve 10–100× lower energy drift than black-box models, confirming that encoding the correct mathematical structure as an inductive bias is critical

Table 1: Model comparison ($t = 10\,\mathrm{s}$ rollout, $\theta_0 = \pi/4, \pi/3$)

| **Model** | **MSE** (2s) | **MSE** (10s) | $\Delta E_{\max}$ |
|---|---|---|---|
| Baseline MLP | $2.3 \times 10^{-2}$ | $4.7 \times 10^{-1}$ | 8.2 |
| ResNet | $6.4 \times 10^{-3}$ | $1.5 \times 10^{-1}$ | 3.1 |
| Neural ODE | $2.5 \times 10^{-3}$ | $9.8 \times 10^{-2}$ | 1.2 |
| HNN | $9.0 \times 10^{-4}$ | $3.2 \times 10^{-2}$ | **0.005** |
| LNN | $9.6 \times 10^{-4}$ | $3.8 \times 10^{-2}$ | 0.008 |
| PINN | $1.6 \times 10^{-3}$ | $5.5 \times 10^{-2}$ | 0.15 |

for physical systems. Neural ODE provides continuous-time dynamics but without inherent energy conservation. PINN effectively constrains the solution space via the physics residual but requires careful loss balancing (we use a curriculum strategy, increasing $\lambda_{\text{phys}}$ linearly over training).

Qualitatively, black-box models produce trajectories that appear reasonable for $t < 2\,\mathrm{s}$ but diverge rapidly at longer horizons. HNN and LNN maintain physically plausible phase-space orbits even at $t = 10\,\mathrm{s}$, though pointwise accuracy degrades due to chaotic divergence.

# 4 Discussion

Our results highlight a key insight: *for chaotic systems, conservation laws matter more than pointwise accuracy.* Due to Lyapunov divergence, no model can track individual trajectories beyond a few characteristic times. However, structure-preserving models maintain qualitatively correct behavior (bounded orbits, energy conservation) indefinitely, while black-box models eventually produce physically meaningless states.

Limitations include: (i) we use the Lagrangian-coordinate state for HNN training, requiring conversion to canonical momenta; (ii) LNN training requires computing second-order derivatives through the network, which is computationally expensive; (iii) PINN's per-trajectory formulation limits its generalization. Future work could explore compositional approaches (e.g., learning separate kinetic and potential energy networks) or extend to dissipative systems.

**Code availability:** `https://github.com/gehaozhang/sciml-double-pendulum`

# References


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